Mean moderation and covariance moderation as data-generating architectures for biomarker-treatment interactions in aggregated N-of-1 trials, and their divergent power under carryover

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1 Abstract

Background. Simulation is the standard tool for power and sample-size calculation in aggregated N-of-1 trials, designs in which each participant is repeatedly crossed on and off treatment and many single-patient series are pooled in order to validate a predictive biomarker. Any such simulation must specify how the biomarker moderates the treatment effect in its data-generating process (DGP). That specification is a substantive choice rather than an implementation detail. We formalize two contrasting parameterizations and show that the choice between them materially changes the estimated power loss under carryover.

Methods. We define two DGP architectures. Under 'mean moderation' the biomarker scales the treatment effect in the conditional mean. Under 'covariance moderation' the interaction is represented as a correlation that depends on drug-exposure condition (on drug versus off drug), in a joint multivariate normal (MVN) distribution. Holding the analysis model fixed at a linear mixed model with continuous-time AR(1) residual correlation, we quantify by Monte Carlo simulation the power to detect the interaction as carryover increases, across three within-subject designs matched to a prazosin and PTSD reference application.

Results. Under mean moderation power is nearly preserved, because carryover does not disrupt the biomarker's relationship to the treatment effect. At the reference cell power falls only from 0.751 to 0.730, a relative loss of 2.8%, as the carryover half-life grows to one week. Under covariance moderation the same carryover erodes the differential correlation that carries the signal, and power falls from 0.777 to 0.539, a relative loss of 30.6%. That is roughly eleven times the mean-moderation loss and far beyond Monte Carlo error ($z = 7.64$, $p < 10^{-13}$). The discrepancy is strongly design-dependent. It is large in designs with a discontinuation phase and negligible in a classical crossover, whose within-subject AR(1) structure already dominates the correlation channel.

Conclusions. The DGP architecture is a substantive scientific assumption rather than a parameterization convenience, and it determines how severely carryover appears to reduce power. Biomarker-validation simu-

lation studies should state which of the two architectures they adopt and, when the mechanism is uncertain, report power under both.

Keywords. N-of-1 trials; biomarker-treatment interaction; data-generating process; multivariate normal; carryover effects; clinical trial simulation; precision medicine; statistical power.

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2 1. Introduction

Precision-medicine trials increasingly ask whether a baseline biomarker modifies the treatment effect, that is, whether a biomarker-treatment interaction is present. For simplified special cases, such as a balanced strict repeated-measures analysis of variance under sphericity, the power of the interaction test does have a closed form, and a companion paper[1] derives it together with its equivalence to a two-sample t-test. For the continuous-time mixed model and the N-of-1 designs considered here, however, no such closed form is available, so power is estimated by Monte Carlo simulation. Many trials are generated and analyzed, and the rejection rate is recorded. The simulation must itself specify the biomarker-treatment interaction in its data-generating process, and there is more than one way to do so.

We distinguish two architectures. The common one, direct mean moderation, adds an interaction term to the outcome equation so that the biomarker scales the size of the treatment effect. Essentially every standard trial-simulation framework uses it[2,3]. The second, covariance moderation through differential correlation, used by Hendrickson et al.[4] for N-of-1 trials, instead makes the biomarker change the correlation between treatment response and outcome, so that the interaction is represented in the joint distribution rather than in the mean. On this reading it is the reduced-form second-moment representation of a latent responder-class mechanism, developed at length in a companion paper[5]. Patients may belong to unobserved responder and non-responder groups, with the biomarker only an imperfect indicator of that status, so that two patients with the same biomarker value can still respond differently. Mean moderation cannot represent this situation, because it assumes that the biomarker fixes each patient's drug effect exactly. Covariance moderation represents it by letting biomarker and response track one another only while the drug is acting, as a tendency rather than a fixed rule, a tendency that fades once the drug washes out. As Section 4 shows, that difference is why carryover costs more power under covariance moderation than under mean moderation. A third, 'combined' architecture activates both channels simultaneously through two independent parameters and recovers these two as its single-channel limits. Because it introduces a second free parameter and its power behavior warrants separate treatment, it is developed in a companion paper[6] and is not considered further here.

Choosing between these two architectures is not a matter of computational convenience, because each carries genuine advantages and a genuine cost. Mean moderation is simple, has a single free parameter, and preserves the biomarker-treatment effect ratio under carryover. It commits, however, to a strong assumption, namely that the biomarker deterministically fixes each patient's drug response, an assumption that is implausible whenever the biomarker is only a statistical proxy for an unobserved responder subtype. Covariance moderation relaxes that assumption. As the reduced-form representation of a latent responder-class mechanism discussed above, it lets the biomarker predict response only probabilistically, which is the more defensible choice whenever responder status, rather than the biomarker itself, is the true causal driver, as is plausibly the case for the noradrenergic-tone hypothesis behind the prazosin and PTSD reference application used throughout this paper. That fidelity to the underlying biology comes at a real cost. Because the covariance-moderation signal lives in a correlation between biomarker and response that is sustained only while the drug is active, any off-drug erosion of drug effect directly erodes the statistical signal itself, producing power losses that this paper shows can run many times larger than under mean moderation. Investigators who adopt covariance moderation on biological grounds should therefore budget for its carryover cost at the design stage, rather than discover it after a trial has already been powered.

Hendrickson et al.[4] chose covariance moderation for exactly this reason when they designed the Monte Carlo simulation program behind the prazosin and PTSD trial. Given that trial's own results, covariance moderation was the logical, biologically appropriate specification, because representing the interaction in the mean would have imposed a deterministic biomarker-response relationship that the trial's biology did not support. Mean moderation, by contrast, was and remains the modal specification across the N-of-1 and broader biomarker-simulation literature (Section 4.4), so Hendrickson's choice was a deliberate departure from convention rather than an oversight. What that departure costs, measured in statistical power under carryover, had not previously been quantified. Quantifying it is the central aim of this paper. We take the covariance-moderation choice as scientifically justified for the prazosin and PTSD application, and we measure precisely the size of the power penalty it exacts once carryover is present, so that investigators reasoning by analogy from that trial can budget for the penalty explicitly.

In a single N-of-1 trial one participant is repeatedly crossed between active treatment and a comparator over a sequence of measurement occasions, so that the within-participant contrast between on-drug and off-drug conditions estimates that participant's individual treatment effect. An *aggregated* N-of-1 trial pools many such series in a mixed model to obtain population-level inference at far smaller sample sizes than a parallel-group design. The feature that distinguishes these designs analytically is *carryover*: residual drug effect that decays over days to weeks after discontinuation and contaminates off-drug occasions. We study three within-subject designs that differ in how on-drug and off-drug occasions are arranged, a traditional crossover (CO), an open-label-plus-blinded-discontinuation design (OL+BDC), and a Hybrid N-of-1 design, all defined in Section 3.

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3 2. Methods

3.1 2.1 Notation and analysis model

Throughout, B_i denotes the biomarker value of participant i, D_{it} the binary on-drug indicator at timepoint t (1 on drug, 0 off drug), and BR_{it} the biological response component of the outcome. These three symbols describe the data-generating-process layer. The analysis-model layer uses a related but distinct continuous drug-exposure variable, `Dbc`, the continuous analogue of D_{it}. It equals 1 during on-drug periods and $\exp(-\lambda t_{sd})$ during off-drug periods, where t_{sd} is time since drug discontinuation and λ is the carryover decay rate, typically derived from the drug's pharmacokinetic half-life as $\lambda = \ln 2/t_{1/2}$. Whereas D_{it} flips abruptly between 0 and 1, `Dbc` captures the smooth shrinkage of residual drug effect during the off-drug period. Code-style identifiers in typewriter font refer to variables in the pmsimstats codebase or to R formula syntax. Mathematical symbols are used otherwise. The analysis model is the same under both architectures, an `nlme::lme` fit with a `corCAR1` continuous-time autocorrelation structure, a random intercept, and a single biomarker-by-drug interaction term, written `bm:Dbc` in R formula syntax. The interaction coefficient on `bm:Dbc` tests whether the biomarker moderates the exposure-decayed drug effect.

3.2 2.2 Two data-generating architectures

3.2.1 2.2.1 Mean moderation

In this architecture the biomarker enters the mean structure of the outcome directly. The treatment effect for participant i at timepoint t is

$$Y_{it} = \mu_0 + \beta_1 \cdot D_{it} \cdot (1 + \beta_{bm} \cdot B_i) + \epsilon_{it},$$

where D_{it} is the on-drug indicator (or a continuous measure of drug exposure), B_i is the participant's biomarker value (typically centered), and β_{bm} is the biomarker moderation parameter. The interaction is explicit, deterministic, and lives in the first moment of the outcome. Participants with higher biomarker values have proportionally larger treatment effects, and this population-level ratio is preserved under any proportional scaling of D_{it}, including the shrinkage that carryover imposes.

This multiplicative form maps directly onto the standard additive regression form for predictive biomarker effects used in the trial methodology literature, specifically the effect-modeling equation of Kent et al.[7], $Y_{it} = \beta_0 + \beta_1 D_{it} + \beta_2 B_i + \beta_3 D_{it} B_i + \epsilon_{it}$, with $\beta_3 = \beta_1 \cdot \beta_{bm}$. The additive form additionally accommodates a prognostic main effect β_2 for the biomarker, which the multiplicative form sets to zero by construction. For power calculations on the interaction coefficient the two forms are interchangeable up to this prognostic-effect distinction, and both fall within mean moderation.

3.2.2 2.2.2 Covariance moderation through differential correlation

In this architecture the biomarker and the biological response (BR) component are jointly drawn from a multivariate normal distribution with a correlation that depends on drug-exposure condition,

$$\begin{pmatrix} B_i \\ BR_{it} \end{pmatrix} \sim \mathrm{MVN}\left(\begin{pmatrix} \mu_B \\ \mu_{BR}(t, D_{it}) \end{pmatrix}, \Sigma(D_{it}) \right), \qquad \mathrm{Cor}(B_i, BR_{it}) = \begin{cases} c_{bm} & D_{it} = 1 \\ c_{bm} \cdot e^{-\lambda t_{sd}} & D_{it} = 0, \end{cases}$$

with the exponential-decay form as the default. The limiting cases $\lambda = 0$ (no decay) and $\lambda \to \infty$ (immediate decay) recover the two no-carryover model variants. The population mean of BR_{it} is the same for all participants with the same treatment history, so the interaction is not in the mean structure. It instead emerges from the conditional distribution. Given $B_i = b$,

$$E[BR_{it} \mid B_i = b, D_{it} = 1] = \mu_{BR}(t) + c_{bm} \cdot \frac{\sigma_{BR}}{\sigma_B} \cdot (b - \mu_B).$$

The interaction here is probabilistic rather than deterministic, lives in the second moment of the joint distribution, and persists with attenuated strength off drug, vanishing only in the limit of complete decay. Carryover in the DGP inflates the BR means during off-drug periods, making them resemble on-drug means. This reduces the differential in the correlation structure between drug-exposure conditions and weakens the signal that the analysis model detects. A third, combined architecture, in which both channels operate at once through independent weights, recovers the two architectures above as its single-channel limits and is the natural framework when the operative mechanism is uncertain. It is developed in a companion paper[6].

3.2.3 2.2.3 Analytical comparison

Consider the expected outcome difference between two participants with biomarker values b_1 and b_2 at the same drug exposure level D. Under mean moderation,

$$E[Y \mid b_1, D] - E[Y \mid b_2, D] = \beta_1 \cdot \beta_{bm} \cdot (b_1 - b_2) \cdot D.$$

This scales linearly with D and therefore shrinks during off-drug periods, when D is replaced by an exposure-decayed value. However, the *ratio* of drug effect between high- and low-biomarker participants is preserved at every level of exposure, so the biomarker-treatment signal contracts in amplitude but does not lose identifiability under carryover. Under covariance moderation the analogous quantity, during off-drug periods with carryover, is

$$E[BR \mid b, D = 0, t_{sd}] = \mu_{BR,\text{off}}(t_{sd}) + c_{bm} \cdot e^{-\lambda t_{sd}} \cdot \frac{\sigma_{BR}}{\sigma_B} \cdot (b - \mu_B).$$

Here the biomarker-dependent component decays exponentially with time off drug. As t_{sd} increases, the off-drug conditional expectation converges to $\mu_{BR,\text{off}}$ for all biomarker values, eliminating the differential signal entirely, in contrast to mean moderation's preserved ratio.

3.3 2.3 Simulation design

We ran identical simulation configurations under both architectures using the pmsimstats-ng framework, matching the DGP parameters as closely as possible. The non-biologic response components (time-varying response, placebo-belief accumulation, and residual noise) are generated from the same multivariate normal draw, with identical means, variances, within-factor AR(1) autocorrelation, and cross-factor correlations, under both architectures. The two DGPs differ only in how the biomarker-to-biological-response linkage is specified, so the divergent carryover behavior reported in Section 3 is attributable to that parameterization alone and not to any difference in how the other components are generated.

We follow the ADEMP framework of Morris, White, and Crowther[8], comprising Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures. The Aim is to compare power to detect the biomarker-by-drug interaction across architectures as carryover increases. The Data-generating mechanisms are the two architectures of Section 2.2. The **primary estimand** is the interaction coefficient $\beta_{bm:D}$ in the model `Sx ~ bm + t + Dbc + bm:Dbc`. The **Methods** setup is as follows.

- Trial designs: CO (traditional crossover), Hybrid (N-of-1 Hybrid), OL+BDC (open-label with blinded discontinuation).
- Sample size: $N = 70$ in total for every design, allocated across each design's randomization paths, so the three designs are matched on N. The two-path CO and OL+BDC designs put 35 on each path, and the four-path Hybrid design splits 70 as 18/18/17/17.
- Biomarker correlation or moderation: $c_{bm} = 0.45$, equivalently $\beta_{bm} = 0.45$.
- DGP carryover half-life: $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks, with the analysis-model `Dbc` half-life matched to it.
- Replicates: 1,000 per cell (Monte Carlo standard error, MCSE, on a binomial power estimate near 0.5 is approximately 0.016).
- Implementation: 8-core `parallel::mclapply`, full factorial of 2 architectures $\times$ 3 designs $\times$ 3 $t_{1/2}$ $\times$ 2 c_{bm} levels gives 36 cells and 36,000 fits, with 100% convergence.

The **Performance measures** are power for $\beta_{bm:D}$ at $\alpha = 0.05$, Type I error under $c_{bm} = 0$ (both with MCSE $\sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}}$), bias of $\hat{\beta}_{bm:D}$ relative to the reference true value, the empirical MCSE of $\hat{\beta}_{bm:D}$ across replicates, and the convergence rate. The number of replicates per cell is chosen to achieve MCSE below 0.02 for power at $\hat{\pi} = 0.5$, which corresponds to $n_{\text{reps}} \geq 625$. We use $n_{\text{reps}} = 1000$ throughout.

The $N = 70$ choice is the smallest of the prazosin-PTSD-matched sample sizes used in the published methodological literature, and it is a regime in which no design or architecture saturates power at the no-carryover cells (the highest cell in the grid is 0.842), so carryover-induced erosion is mechanically visible. This is the main practical gain from matching the designs on N. At a previous per-path parameterization the Hybrid cells sat above 0.99, where no erosion could register.

4 3. Results

Covariance moderation, as Section 1 argues, is often the more biologically defensible choice when a biomarker is best understood as predictive rather than causal, and it is the specification Hendrickson et al. selected for the prazosin and PTSD application on exactly these grounds. That defensibility does not eliminate a carryover cost. Quantifying that cost, so that the appeal of the architecture does not obscure a real design consideration, is the purpose of what follows.

4.1 3.1 Power under carryover at $N = 70$

Mean moderation.

Design	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
CO	0.739	0.739	0.725
Hybrid	0.842	0.844	0.784
OL+BDC	0.751	0.748	0.730

Power declines modestly under mean moderation in the two-period designs and more appreciably in the Hybrid design. The relative loss from $t_{1/2} = 0$ to $t_{1/2} = 1$ is 1.9% (CO), 6.9% (Hybrid), and 2.8% (OL+BDC). Only the Hybrid loss is separable from zero at 1{,}000 replicates per cell ($z = 3.34$; CO $z = 0.71$, OL+BDC $z = 1.07$). All three values fall at or below the 8-13% relative-loss range suggested by earlier exploratory runs in this project for direct-mean-moderation DGPs, an erosion that leaves power substantially intact, though the Hybrid design shows it is not uniformly negligible.

Covariance moderation (differential correlation, MVN).

Design	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
CO	0.745	0.754	0.723
Hybrid	0.840	0.795	0.711
OL+BDC	0.777	0.694	0.539

Power declines more sharply under covariance moderation, most markedly in the OL+BDC design. The relative loss from $t_{1/2} = 0$ to $t_{1/2} = 1$ is 3.0% (CO, not significantly different from mean moderation's CO loss), 15.4% (Hybrid), and **30.6% (OL+BDC)**, roughly 11 times the matched mean-moderation OL+BDC loss (2.8%). This is the power cost of the more biologically faithful specification, and it is largest in exactly the design, OL+BDC, with the longest blinded off-drug window. The CO cell shows a small, non-monotone fluctuation ($0.745 \to 0.754 \to 0.723$), within one MCSE and consistent with no architecture-by-carryover interaction in that design. These figures fall below the 40-60% relative-loss range suggested by earlier exploratory runs for the MVN differential-correlation DGP, but confirm their qualitative finding of substantial, design-dependent erosion.

We compare the carryover-induced power loss between architectures using a difference-of-differences z-statistic. Let $\Delta_X = \hat{\pi}_{X,0} - \hat{\pi}_{X,1}$ denote the within-architecture loss from $t_{1/2} = 0$ to $t_{1/2} = 1.0$ for architecture X, estimated from $n = 1{,}000$ independent replicates per cell. The covariance-minus-mean gap $\Delta_B - \Delta_A$ has standard error

$$\mathrm{SE} = \sqrt{\frac{\hat{\pi}_{B,0}(1-\hat{\pi}_{B,0})}{n} + \frac{\hat{\pi}_{B,1}(1-\hat{\pi}_{B,1})}{n} + \frac{\hat{\pi}_{A,0}(1-\hat{\pi}_{A,0})}{n} + \frac{\hat{\pi}_{A,1}(1-\hat{\pi}_{A,1})}{n}},$$

since the four proportions come from independent replicate sets (all tests two-sided). At OL+BDC, $\Delta_A = 0.751 - 0.730 = 0.021$ and $\Delta_B = 0.777 - 0.539 = 0.238$, giving $\Delta_B - \Delta_A = 0.217$ (about 22 percentage points) and $z = 7.64$ ($p < 10^{-13}$). The architectures' carryover behavior differs well beyond Monte Carlo error. The same contrast is positive but smaller for Hybrid ($\Delta_B - \Delta_A = 0.071$, $z = 2.80$, $p = 0.005$) and indistinguishable from zero for CO ($z = 0.29$), because the CO design's within-subject AR(1) structure dominates the carryover-mediated correlation-decay channel that covariance moderation uses to represent the interaction. The ordering of the three designs on this contrast, OL+BDC well above Hybrid and Hybrid above CO, is the substantive result. It is read at a common $N = 70$ and is not a statement about the designs' relative power.

4.2 3.2 Mechanism: why the architectures diverge

Under mean moderation, carryover reduces the magnitude of the drug exposure variable seen by the analysis model (`Dbc` drops below 1 off drug), and the mean BR at off-drug timepoints is inflated by the residual drug effect, exactly as under covariance moderation. Both mechanisms compress the treatment contrast between drug-exposure conditions and are shared by the two architectures. What mean moderation lacks is a second channel of signal loss. β_{bm} is a fixed population parameter, the biomarker enters the mean structure directly rather than through a correlation, and the noise structure is independent of drug-exposure condition. The interaction term therefore has reduced variance (`Dbc` has less contrast between drug-exposure conditions), but its signal-to-noise ratio degrades only modestly.

Under covariance moderation, carryover operates on two levels simultaneously. **Mean blurring**: off-drug BR means are inflated by residual carryover, resembling on-drug BR means and reducing the treatment contrast. This channel is shared with mean moderation. **Correlation erosion**: the BM-BR correlation decays exponentially with t_{sd} off drug, a structural consequence of the DGP rather than a modeling artifact, because the correlation reflects the degree to which the drug effect is active. As the drug washes out, the drug-mediated component of BR variance shrinks relative to the drug-independent component, so the correlation weakens. This channel is unique to covariance moderation, because only there does the biomarker signal live in the second moment of the joint distribution. Under mean moderation the drug-mediated and drug-independent components of BR variance move in tandem, so their ratio, and hence the identifiability of the interaction, is preserved even as amplitudes shrink. The `bm:Dbc` coefficient depends on the covariance between B_i and `Dbc`-weighted outcomes. Under covariance moderation this covariance is attacked from both sides at once, while mean moderation escapes the second attack entirely, which accounts for the divergence in power loss documented above.

4.3 3.3 Robustness check at $N = 140$

Section 3.1 puts $N = 70$ on every design, the regime in which carryover-induced loss is mechanically visible because no architecture saturates power. We conducted a confirmation run at $N = 140$ to check whether the qualitative ordering survives the power saturation that comes with a larger sample, using an 800-replicate-per-cell Monte Carlo run across 12 cells (architecture $\in \{\text{MVN}, \text{mean moderation}\} \times c_{bm} \in \{0, 0.45\} \times t_{1/2} \in \{0, 0.5, 1.0\}$ weeks), all on the OL+BDC design (100% convergence across 9{,}600 fits).

Architecture ($c_{bm} = 0.45$)	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
Covariance moderation (MVN)	0.978	0.945	0.885
Mean moderation	0.976	0.973	0.959

The qualitative ordering matches Section 3.1. Covariance moderation loses 9.3 percentage points ($z = 7.4$, $p < 10^{-12}$) against 1.7 for mean moderation ($z = 2.0$, $p \approx 0.05$). The absolute magnitudes are compressed because $N = 140$ places both architectures within 2-5 percentage points of the unit ceiling at $t_{1/2} = 0$, mechanically truncating the loss either curve can absorb (3% and 9% relative loss for covariance moderation versus 0.4% and 1.8% for mean moderation at this N). The production results in Section 3.1 remain the canonical reference for the narrative magnitude. This check confirms only that the ordering is not an artifact of the smaller sample.

4.4 3.4 Type I error and estimator bias

Across all 18 null cells of the Section 3.1 production run, Type I error sat in $[0.010, 0.074]$, with mean 0.042 and median 0.035, and no architecture-specific inflation. The uniformly sub-nominal rates are consistent with the `corCAR1` working-covariance miscalibration characterized in a companion calibration study[9], whose model-based standard error is over-stated by roughly 6 to 10 percent. Because this affects both architectures alike, it is common-mode with respect to the architecture contrast and does not affect it.

Mean estimated $\hat{\beta}_{bm:D_{bc}}$ at $c_{bm} = 0.45$ ranged from -0.230 to -0.281 across the 18 alternative cells, with the mean-moderation OL+BDC cell at $t_{1/2} = 1.0$ producing the largest absolute mean. Covariance-moderation cells maintained a mean near -0.234 across all $t_{1/2}$ values, consistent with that architecture's information channel being eroded through correlation decay rather than coefficient drift. Empirical SE of $\hat{\beta}_{bm:D_{bc}}$ runs 0.07 to

0.10 across cells, and the within-replicate model SE matches it at scale, which is consistent with nominal-95% Wald coverage but was not directly verified by a coverage simulation.

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5 4. Discussion

5.1 4.1 Matching architecture to biological mechanism

The choice of architecture is a choice about *why* the biomarker predicts response, and is therefore a biological statement rather than a coding preference. Mean moderation is appropriate when the biomarker mechanically sets the size of the drug effect, for example by fixing effective dose or receptor occupancy, so that doubling the biomarker roughly doubles the effect at every timepoint. Because carryover keeps the same proportion off drug, power is barely touched. Typical examples include drug clearance (kidney function setting the steady-state level of a renally cleared drug), receptor density, and metabolizer genotype. Covariance moderation is appropriate when the biomarker is only a proxy for a hidden responder status that it indexes imperfectly, so that identical biomarker values can still respond differently, and the biomarker predicts response only while the drug is acting. Once the drug washes out, the correlation that carried the signal fades and power drops sharply, as in Section 3. Typical examples include resting blood pressure as a marker of noradrenergic tone, inflammatory markers such as CRP or IL-6, and polygenic risk scores. A fully faithful model of the hidden responder-class idea would be a finite mixture. Covariance moderation is a convenient correlation-level approximation, adequate for power calculations but not a claim about the true mechanism. A companion paper[5] develops the mixture and latent-class formulations, deriving when a single multivariate normal reproduces a two-component mixture's first two moments and cataloguing where it fails. Senn[10] also cautions that apparent biomarker-treatment effects are sometimes just variance artifacts, so covariance moderation's signal may be smaller than assumed.

The pmsimstats software was built for a prazosin/PTSD trial[4,11], in which the biomarker, resting blood pressure, is thought to be a proxy for noradrenergic tone rather than a direct driver of drug levels, a covariance-moderation story. The carryover-driven power loss we find under covariance moderation is therefore a clinically relevant warning. Designs with long off-drug periods may lose more power to detect this biomarker than a mean-moderation simulation would suggest. Should blood pressure act through the mean channel as well as the correlation channel, the dual-channel architecture of the companion paper[6] is the appropriate model, and because it estimates two independent parameters, it calls for independent evidence on each.

Three further considerations support covariance moderation as the more defensible specification for this application. First, resting blood pressure is a single-occasion vital sign subject to substantial physiological and measurement variability, so treating it as a precise multiplicative scale factor on the treatment effect assumes a level of measurement precision the biomarker does not have. A specification that asserts only an association, and its strength, is proportionate to that measurement error. Second, the distinction tracks an established one in the predictive-biomarker literature. The PATH framework of Kent et al.[7] separates a causal treatment effect modifier from a biomarker that is merely predictive of response without being causally upstream of it. Covariance moderation is the natural representation of the predictive role and mean moderation of the causal-mediator role, so classifying blood pressure as predictive rather than causal in this trial argues for covariance moderation on grounds already established in that literature. Third, mean moderation requires committing to a specific functional form for how the biomarker scales the treatment effect, typically linear in the centered biomarker. Covariance moderation makes the weaker assumption that biomarker and response are associated, without specifying the functional form of that association beyond its strength, which is the more parsimonious choice when investigators lack direct evidence for a particular dose-response shape.

5.2 4.2 Reporting recommendations

The choice of DGP architecture should be guided by the hypothesized biological mechanism, as summarized above. When the mechanism is

Criterion	Mean moderation	Covariance moderation
Biomarker role	Causal mediator	Statistical predictor
Mechanism	Deterministic scaling	Probabilistic association
Signal location	Mean structure	Covariance structure
Carryover impact	Modest (up to 7%)	Design-dependent, up to 31%
Appropriate when	Biomarker determines dose or PK	Biomarker indexes a latent subtype

ambiguous, as it often is in early biomarker development, we recommend that investigators run the simulation under both architectures and report the range of power estimates, treating covariance moderation as the conservative default because its larger carryover sensitivity keeps power estimates from being optimistic. A dual-channel sensitivity sweep across mixing weights is available in the companion paper[6]. Designing the trial to minimize carryover, through adequate washout periods, also reduces sensitivity to the architecture choice. More generally, simulation studies evaluating biomarker-moderated treatment effects should state explicitly the following.

- Whether the interaction is generated through mean moderation, covariance moderation, or both channels simultaneously.
- The biological rationale for the chosen mechanism.
- Whether any carryover model operates on the interaction signal consistently with that choice.
- When the mechanism is uncertain, sensitivity analyses under the alternative architecture.

These requirements specialize the Data-generating element of the ADEMP framework[8] to the class of architecture-sensitive biomarker simulations.

5.3 4.3 Possible mitigation strategies

Under covariance moderation, the power loss has two sources. The drug-exposure variable's variance shrinks (recoverable), and the biomarker-response correlation genuinely fades off drug (lost information no model can recover). Because off-drug points carry less signal, methods that down-weight or drop them may recover part of the loss. Candidates include restricting the analysis to on-drug points, weighting observations by expected residual drug effect (via the `weights` argument in `nlme::lme`), a within-subject contrast regressing each patient's on-minus-off mean difference on the biomarker, a two-stage random-slopes approach, or excluding the most contaminated early off-drug points. None of these strategies was tested by simulation here. A companion paper on carryover-mitigation strategies takes that up. We regard a systematic simulation comparison, analogous to the factorial comparison of Section 3, as important future work, and expect the on-drug-only restriction and the weighted analysis to be the most tractable candidates for a first empirical evaluation.

5.4 4.4 Relation to the wider literature

Almost every trial-simulation framework uses mean moderation. Simon[2], Freidlin and Korn[3], Renfro and Sargent[12], and standard N-of-1 simulation studies all put the biomarker effect in the mean. Even methodological work on N-of-1 carryover sharing authors with Hendrickson et al.[4] uses mean moderation, which suggests Hendrickson's differential-correlation choice was a one-off rather than a convention. The practical worry is that most published power estimates may be optimistic for biomarkers that really behave like covariance moderation, because they ignore the extra carryover-driven loss demonstrated here. The distinction matters most exactly in the designs precision medicine favors: treatment-switching designs (crossover, N-of-1 hybrid, basket/platform trials) put patients through off-drug phases where the covariance-moderation correlation decays, so the interaction signal in later periods is weaker than in the first. Parallel-group enrichment designs, with no off-drug phase, are immune, and give similar power under either architecture. Adaptive enrichment trials that use interim power calculations written in the mean-moderation idiom are at particular risk. If the truth is closer to covariance moderation and patients pass through washouts, the real power is lower than the rule assumes, which can trigger premature stopping or an over-narrow eligible population. This is a data-generating-process question, separate from, and not answered by, the analysis-model guidance in the PATH statement[7]. The architecture choice matters most in exactly the within-subject designs precision medicine finds most attractive, and least in the parallel-group designs where it is often, wrongly, assumed to be harmless.

5.5 4.5 Limitations

Several limitations qualify these results. The magnitude of the covariance-moderation loss reported here (up to 30.6%) falls below the 40-60% range suggested by earlier exploratory runs in this project, so the precise figure should be read as design- and calibration-specific rather than universal. The uniformly sub-nominal Type I error we observe reflects a known conservatism of the `corCAR1` Wald test that is common-mode across architectures but not itself resolved here. A cluster-robust remedy is characterized in a companion calibration study[9]. Nominal coverage was inferred from the correspondence between empirical and model-based standard errors rather than verified by a dedicated coverage simulation. The mitigation strategies of Section 4.3 are mechanistic hypotheses, not simulation-tested recommendations. Finally, the combined architecture that activates both channels at once is deferred entirely to a companion paper[6], and our results are matched to a single reference application (prazosin/PTSD) and three within-subject designs. Generalization to other biomarkers, designs, or effect sizes should be checked rather than assumed.

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6 5. Conclusions

1. The choice between direct mean moderation and covariance moderation for generating biomarker-treatment interactions in clinical trial simulations has substantive consequences for power estimates under carryover.

2. Covariance moderation is intuitively appealing whenever a biomarker is better understood as predictive than causal, and Section 4.1 sets out the reasons this is often the more defensible specification. That appeal carries a measurable cost under carryover. Under mean moderation, carryover reduces power only modestly across the three designs examined (1.9% to 6.9% relative), because the proportional biomarker-drug relationship is preserved off drug. Under covariance moderation the loss is strongly design-dependent, from 3.0% (crossover, not distinguish-

able from zero) to 15.4% (Hybrid) to 30.6% (OL+BDC), because the differential-correlation signal erodes as the drug effect wanes.

3. The choice should be guided by the hypothesized biological mechanism. Mean moderation is appropriate for biomarkers that directly determine drug pharmacokinetics or pharmacodynamics, covariance moderation for biomarkers that statistically predict treatment response through latent subtype membership.

4. For the prazosin-PTSD application, with blood pressure as the predictive biomarker, covariance moderation is the more appropriate model, and the resulting power sensitivity to carryover should inform trial-design decisions about off-drug period duration.

5. The existing clinical trial simulation literature uses mean moderation almost exclusively. The Hendrickson et al. (2020) MVN approach is, to our knowledge, unique in the N-of-1 context. This means published power estimates for biomarker-moderated crossover and N-of-1 designs may be systematically optimistic for biomarkers that operate through a correlation-based mechanism.

6. Simulation studies for biomarker-moderated treatment effects should explicitly report their DGP architecture and, when the biological mechanism is uncertain, consider sensitivity analyses under the alternative.

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7 Reproducibility

N denotes the total number of participants across all randomization paths, and Section 3.1 runs every design at the same total, $N = 70$, allocated across that design's paths, so the cross-design comparison in Section 3 is a design comparison rather than a sample-size comparison. An earlier version of this analysis fixed the per-path count at 35 instead of the total, which gave the four-path Hybrid design 140 participants against 70 for the two-path designs and confounded any Hybrid advantage with twice the sample. Section 3.1 has been re-run at matched totals, and only the Hybrid figures changed. Section 3.3 is a deliberate sample-size check on the OL+BDC design alone, at $N = 140$, and is unaffected by this matching since it makes no cross-design comparison. A structured, section-by-section summary of the argument, without the derivations, tables, or citations, is provided as a supplementary document (`bullets.Rmd`).

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